

- 1 The photograph shows a zebrafish, *Danio rerio*.
Zebrafish live in freshwater throughout South Asia.



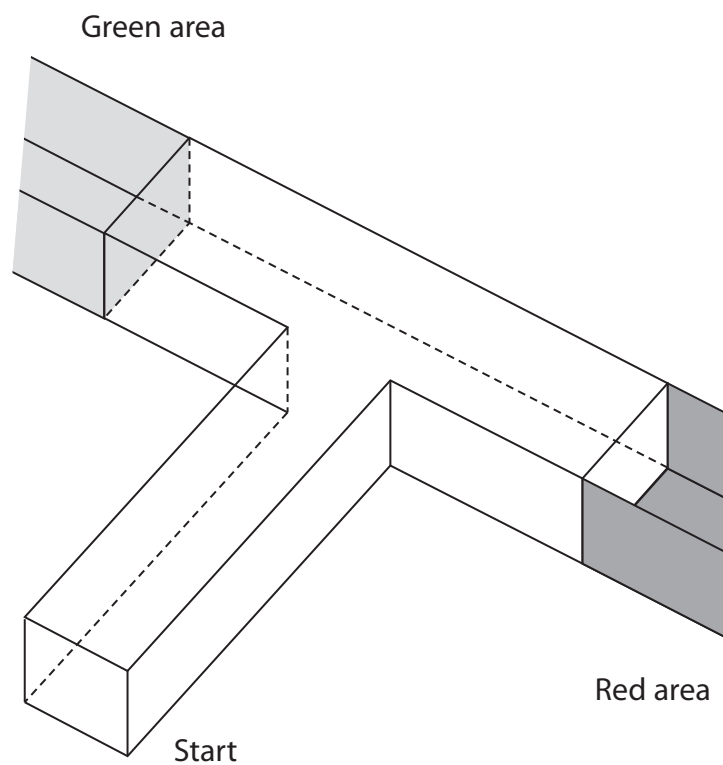
Magnification $\times 1$

(Source: © Pablo Bou Mira/Alamy Stock Photo)

Zebrafish are social animals. They can be kept in large numbers in tanks in the laboratory.

A scientist investigated the preference of zebrafish for different coloured areas of a T-maze, filled with water.

The diagram shows the T-maze used in this investigation.



A group of 72 zebrafish was used in this investigation.

These fish were raised from eggs of zebrafish living in a river.

Each fish was placed at the start of the T-maze and the colour area the fish swam to was recorded.

The results are shown in the table.

Number of fish preferring	
Green area	Red area
44	28

- (a) Each fish was returned to the source that the eggs came from.

Suggest **one** reason why the scientist thought it was acceptable to carry out this investigation.

(1)

.....

.....

.....

- (b) (i) The scientist made a prediction:

There is no difference between the observed (O) and expected (E) colour preference of the zebrafish.

Using the formula calculate the value of Chi squared.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

(3)

Answer

(ii) The critical value of Chi squared was 3.84.

Give a conclusion that could be made from this investigation.

(1)

(c) (i) State **one** abiotic variable that could affect the results of this investigation.

(1)

Abiotic variable

(ii) Describe how this abiotic variable could be controlled and the effect it could have on the results if it is not controlled.

(2)

Variable

Describe how this variable is controlled

Describe the effect it could have on the results if it is not controlled.

(d) Suggest why the scientist used each zebrafish only once.

(1)

(Total for Question 1 = 9 marks)